

Ethereum Order-Flow Alpha

Short-horizon return prediction and a cost-aware trading strategy
Binance USD-M ETHUSDT perpetual · 2023-03-01 → 2026-09-30

1. Raw data Binance archive 1,310 daily trade files, book depth, funding, OI SHA-256 verified	2. Quality control parse, ms/μs detect, sort, dedupe, gaps, invalid values, frozen book feed detection	3. 1-minute bars 1.88 bn trades → 1,886,400 bars with taker buy/sell flow + as-of book join	4. Features + targets 47 causal features 5 groups; forward returns 5/15/30/60 m leakage audit
5. Statistics HAC regressions, deciles + block bootstrap, events, Holm / BH	6. Walk-forward ML logit / RF / LightGBM 7 folds, purge + embargo, ablation A→E, tuning	7. Freeze → holdout methodology frozen + hash-logged, then one evaluation on 2025-10 → 2026-09	8. Cost-aware backtest next-bar VWAP, fees, slippage, funding, break-even cost, robustness, SOL

Answer in one line. The 1-minute ETH microstructure contains **real, stable, out-of-sample** predictability (holdout AUC **0.5436**, 95% CI 0.5394–0.5480), but the edge is worth only **0.49 bps per side** per trade against a realistic **5.52 bps** taker cost. The strategy earns a gross Sharpe of 2.03 (+67.5%) but loses **-99.6%** after costs: **statistically significant, economically untradable**.

Generated 2026-10-07 by scripts/make_pdf_report.py. Every table is read from the pipeline output files at build time. Code, configuration and all result tables: github.com/kudratikhoda12-design/KK (branch `claude/brave-tesla-wpvjj7`).

1. Research question and design

Question. Does order-flow and market-microstructure information contain statistically significant predictive information about short-horizon Ethereum returns, and does that information translate into economically meaningful trading alpha after transaction costs, slippage and funding?

The project separates three claims that are often conflated, and tests each with its own tool:

- **Statistical predictability:** HAC regressions, decile spreads, event study, multiple-testing control.
- **Out-of-sample predictability:** leakage-controlled walk-forward ML, then one evaluation on a frozen, untouched year.
- **Economic predictability:** a cost-aware backtest with realistic execution, and the break-even cost.

Time split (fixed in Stage 1, before any return was looked at)

Segment	Period	Use
Development	2023-03-01 → 2025-09-30 (31 months)	EDA, statistics, features, walk-forward CV, model and threshold selection
Holdout	2025-10-01 → 2026-09-30 (12 months)	Untouched until the methodology was frozen; evaluated once
Replication	SOLUSD, same dates	Same frozen settings, nothing re-tuned

Timeline of one decision

Step	Definition
Features	data stamped before the close of bar t (trades of bars $\leq t$, last book snapshot $<$ close, settled funding, OI lagged ≥ 5 min)
Signal	at the close of bar t
Execution	VWAP of bar $t+1$ (never the price that generated the signal)
Holding	15 bars per signal; staggered, net position = mean of the last 15 signals ($ \text{position} \leq 1$, no leverage)
Target	$(P[t+15] - P[t]) / P[t]$, P = last trade price at or before the bar close; binary label up = $1\{\text{return} > 0\}$

2. Raw data → research dataset

Source. Binance public archive (data.binance.vision), USD-M futures. Every file is checked against Binance's published SHA-256 before use, and logged in a manifest (URL, bytes, expected and actual hash). Raw files are streamed one day at a time, because one month of trades needs more than 13 GB of RAM.

Dataset	Fields	Volume
aggTrades (daily)	agg_trade_id, price, quantity, first/last trade id, transact_time (ms), is_buyer_maker	1,310 files, 25.3 GB zipped
bookDepth (daily)	timestamp, percentage ($\pm 1...5$), depth, notional: cumulative resting quantity within k% of price	1,310 files, ~30 s snapshots
fundingRate (monthly)	calc_time, funding_interval_hours, last_funding_rate	43 files
metrics (daily)	open interest and long/short ratios, 5-minute	1,310 files
bookTicker (3 sample days)	best bid / ask: used only to measure the quoted spread	archive ends 2024-03-30

Trade direction is given, not inferred. is_buyer_maker = true means the seller was the aggressor, so taker buy volume = trades with is_buyer_maker = false. No tick rule or Lee-Ready classification is needed. **The order book is percentage-band depth, not level-by-level L2:** no best bid/ask, spread or queue.

Data-quality results (full report: docs/data_quality_report.md)

- Raw aggregated trades read: 1,877,097,488 (from 1,310 daily files)
- Removed as data errors: 0 (null/non-finite 0; price ≤ 0 0; qty ≤ 0 0; first_id > last_id 0; outside file's day 0; bad side flag 0)
- Duplicates: exact rows 0; conflicting same-id rows 0
- Bars written: 1,886,400; expected minutes in period: 1,886,400; duplicate timestamps: 0; sorted: True
- Bars with ≥ 1 trade: 1,886,347; minutes without any trade: 53 (0.0028%) in 4 runs; runs ≥ 2 min: 3
- Snapshots kept: 3,710,848 (expected $\approx 3,772,800$ at 2880/day = 98.358%)
- Frozen feed: 97,477 snapshots repeat all 20 band values of their predecessor (2.63%); the longest run is 94,730 snapshots (2025-04-16 10:31:05+00:00 → 2025-05-19 10:26:32+00:00). Book age is measured from the last *changed* snapshot, so these minutes are flagged book_stale (docs/stage2/book_frozen_runs.csv).
- Short/missing days (< 90% of 2880 snapshots), flagged book_short_day in the bars and not interpolated: 9

Key cleaning decisions

Treatment	Detail
Removed (counted)	nulls, price ≤ 0 , qty ≤ 0 , first_id > last_id, timestamp outside the file's day, exact duplicates, conflicting duplicate ids: 0 rows met any rule
Flagged, kept	83 extreme 1-minute moves (> 3%), all liquidation cascades with tens of thousands of trades; bad-tick candidates; trade-id gaps; empty minutes
Frozen book feed	2025-04-16 → 2025-05-19: the archive repeats one snapshot 94,730 times. Book age is measured from the last <i>changed</i> snapshot, so these minutes are stale and the book features are missing
Schema change	$\pm 0.2\%$ bands added on 2026-01-15 (inside the holdout): not used
Other	91 zero open-interest readings and $\pm \text{inf}$ ratios set to missing; nothing interpolated or winsorised
Verification	time zone and band sign checked against trade prices: 94.1% of snapshots have bid-band avg < VWAP < ask-band avg at zero shift vs 64% at ± 1 h

3. Feature engineering and leakage control

47 features are computed on the complete 1-minute grid with **trailing windows only**. Ablation sets are cumulative: A = price, B = A + volume, C = B + trade flow, D = C + book, E = D + funding/OI.

Group	n	Features
price	16	ret_1, ret_5, ret_15, ret_30, ret_60, vol_15, vol_60, vol_240, vol_1440, mom_240_riskadj, dist_ma_60, dist_ma_240, range_15, dist_vwap_15, hour_sin, hour_cos
volume	6	log_volume_1, volume_z_1440, volume_ratio_15, volume_ratio_60, trade_intensity_15, trade_count_z_1440
flow	13	ofi_1, ofi_5, ofi_15, ofi_60, trade_imb_1, trade_imb_15, signed_flow_15, ofi_15_z_1440, ofi_1_z_1440, avg_trade_size_15, avg_trade_size_z_1440, large_share_15, large_ofi_15
book	7	obi_1pct, obi_2pct, obi_3pct, obi_5pct, obi_1pct_chg_15, obi_1pct_mean_15, depth_1pct_z_1440
funding_oi	5	funding_rate_last, funding_rate_mean_3, minutes_to_funding, oi_chg_60, oi_chg_1440

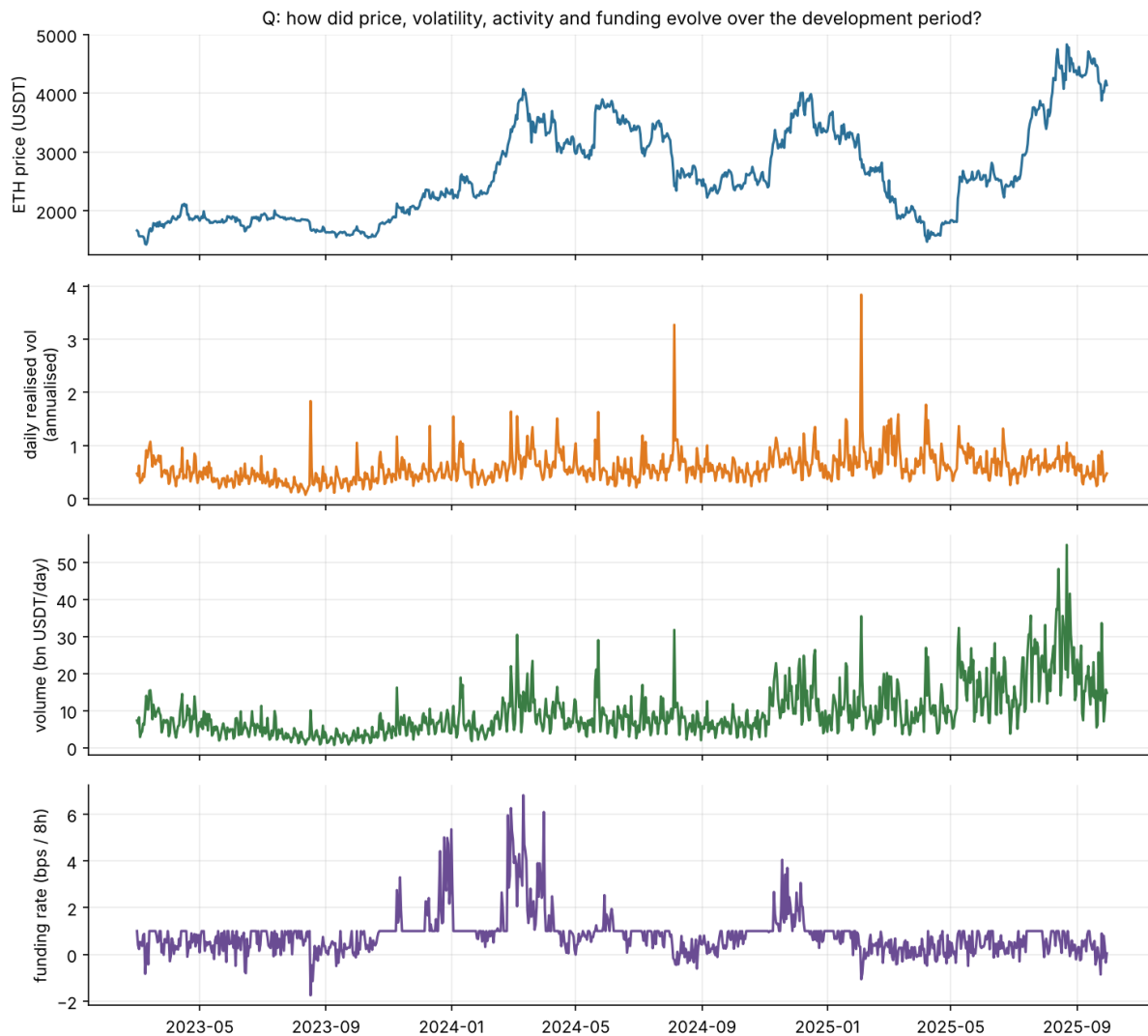
Key definitions: OFI = (taker buy volume – taker sell volume) / total volume over the window; book imbalance at k% = (bid depth within k% – ask depth within k%) / their sum; large trade = aggTrade ≥ 10 ETH (the bucket closest to the median per-minute 95th-percentile trade size in the initial window). Undefined ratios become missing, never zero or ±inf.

Automated leakage audit (scripts/leakage_audit.py)

- PASS: Truncation invariance of all features
- PASS: Book snapshots joined strictly before bar close
- PASS: Funding rate known at decision time
- PASS: Open interest lagged at least 5 minutes
- PASS: Targets are forward returns from the decision price
- PASS: No feature mirrors a target
- PASS: Walk-forward purge / embargo / holdout separation
- PASS: Learned preprocessing fitted inside each training fold

The strongest test is **truncation invariance**: every feature was recomputed on data cut at 12 random times, and all 34,774,174 compared values were identical. Any feature using the future would change when the future is removed.

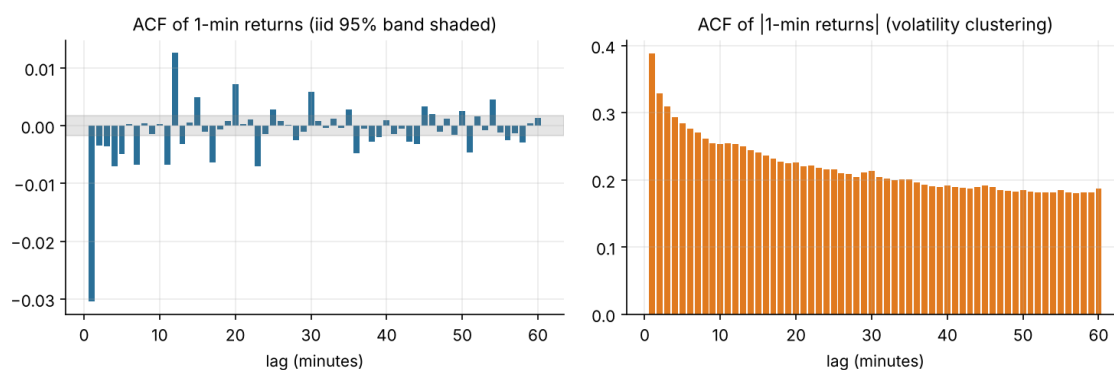
4. Exploratory analysis (development period only)

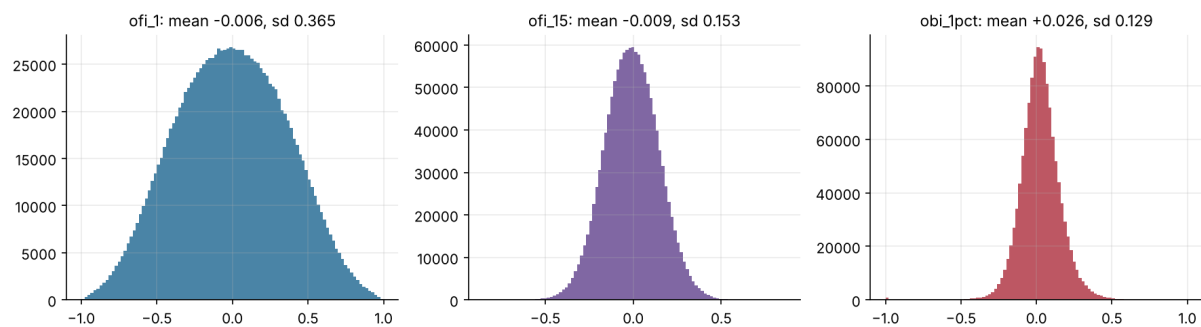


Price, realised volatility, volume and funding over the development period.

Series	n	mean (bps)	sd (bps)	skew	excess kurt.	p1 (bps)	p99 (bps)
1m log return	1,360,739	7.0e-03	8.92	-3.02	487.38	-23.64	23.68
15m forward return	1,360,740	0.16	32.88	-0.67	59.21	-92.04	91.86

Returns are extremely heavy-tailed (1-minute excess kurtosis ≈ 487), which rules out iid-normal inference. 1-minute returns show a small negative first-order autocorrelation, and absolute returns show strong volatility clustering:





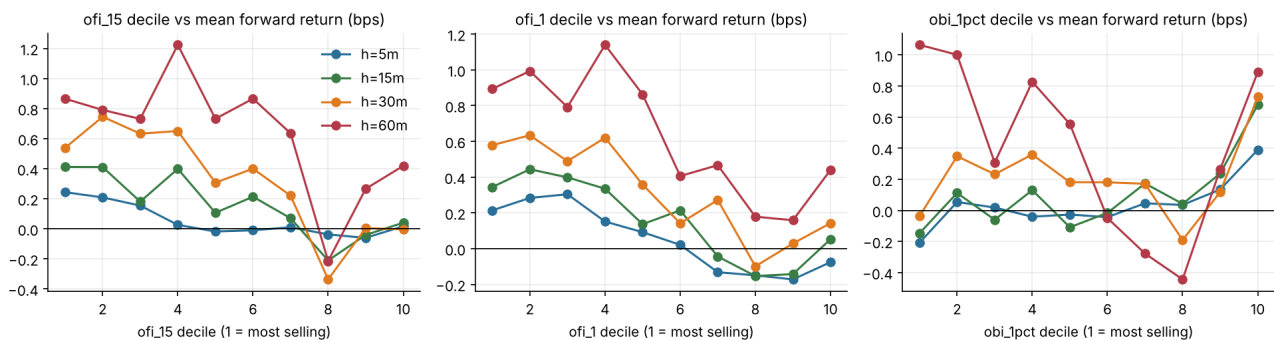
Distributions of 1-minute and 15-minute order-flow imbalance and of the 1% book-band imbalance.

5. Statistical alpha tests (before any ML)

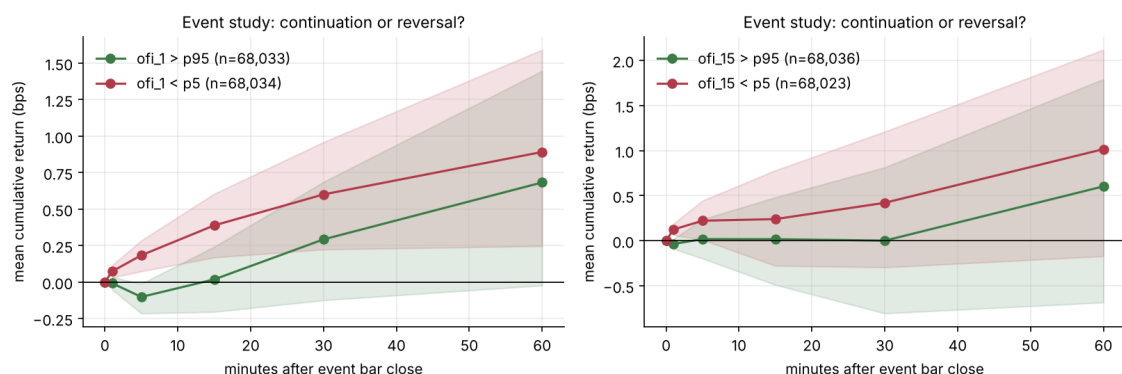
Order flow explains the **same** minute's return very well: $\text{corr}(\text{OFI}_1, r_1) = \mathbf{0.467}$. The question is whether it predicts the **next** minutes. Forward returns overlap, so regressions use Newey–West HAC errors (2h lags) plus a non-overlapping re-estimate. All 36 signal $\times$ horizon tests are Holm and BH adjusted.

Signal	h (min)	Pearson	β (bps / sd)	HAC t	Holm p	non-overlap t
ofi_1	5	-0.008	-0.146	-9.739	7.4e-21	-1.986
ofi_1	15	-0.005	-0.170	-5.990	7.2e-08	-1.094
ofi_1	30	-0.004	-0.194	-4.765	6.2e-05	-2.539
ofi_1	60	-0.004	-0.237	-4.015	0.002	-1.066
ofi_15	5	-0.004	-0.081	-2.820	0.130	-1.513
ofi_15	15	-0.004	-0.148	-2.017	0.961	-3.538
ofi_15	30	-0.005	-0.244	-2.055	0.917	-2.419
ofi_15	60	-0.003	-0.203	-1.140	1.000	0.557
trade_imb_15	5	-0.003	-0.051	-2.499	0.324	-0.958
trade_imb_15	15	-0.003	-0.086	-1.572	1.000	-2.668
trade_imb_15	30	-0.004	-0.178	-1.894	1.000	-1.737
trade_imb_15	60	-0.003	-0.214	-1.357	1.000	0.110
obi_1pct	5	0.008	0.146	4.027	0.002	3.288
obi_1pct	15	0.007	0.237	2.406	0.404	1.223
obi_1pct	30	0.004	0.180	0.979	1.000	0.313
obi_1pct	60	-0.001	-0.084	-0.250	1.000	-0.560

1-minute OFI predicts a small **reversal** at every horizon (Holm-significant), and it survives controls for momentum, volatility and volume. 15-minute OFI is not significant after adjustment. Book imbalance is significant only at 5 minutes and loses significance once recent returns are controlled for.



Mean forward return by signal decile. Effects are fractions of a basis point.



Event study: extreme order-flow minutes are followed by partial reversal, with 95% day-bootstrap bands.

Statistical $\neq$ economic. The effects are highly significant because there are 1.36 million observations, but they are 0.1–0.9 bps, while one round trip costs about 11 bps.

6. Machine learning with walk-forward validation

Validation design. Expanding walk-forward: first 12 months for initial training, then 7 validation blocks of 3 months (2024-03 → 2025-09). Training labels are purged so they end before validation start – 1 day (embargo). Training uses every 5th minute (every 15th for RF), because overlapping labels make neighbours near-duplicates. Imputation and scaling are fitted inside each training fold. Random cross-validation is not used: it would leak shared future price paths.

LightGBM grid (8 configurations, chosen by mean validation log loss)

id	leaves	min leaf	trees	val. log loss	mean AUC	min fold AUC
0	15	500	200	0.6911	0.5388	0.5284
2	15	5,000	200	0.6912	0.5385	0.5277
6	63	5,000	200	0.6912	0.5382	0.5282
1	15	500	600	0.6921	0.5371	0.5279
3	15	5,000	600	0.6921	0.5362	0.5264
4	63	500	200	0.6922	0.5359	0.5269
7	63	5,000	600	0.6925	0.5349	0.5236
5	63	500	600	0.6954	0.5322	0.5242

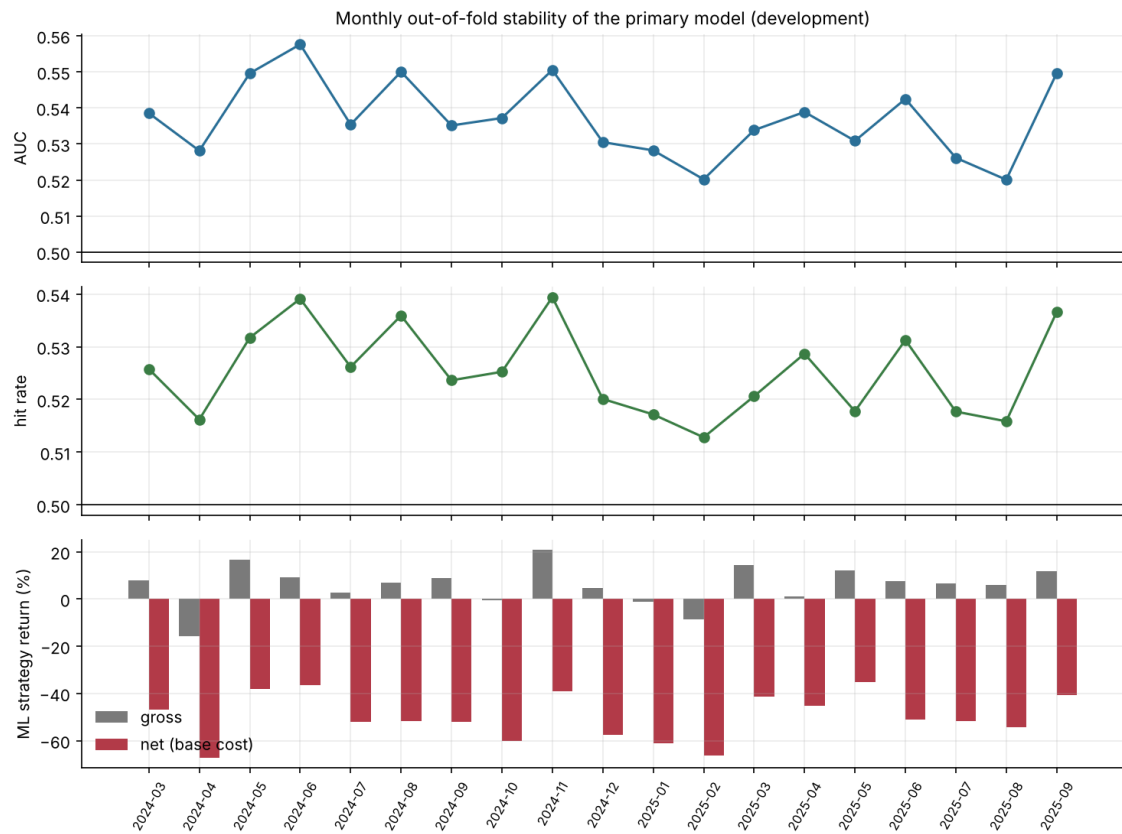
Feature ablation, 15-minute horizon (development, mean over 7 folds)

Model	Set	AUC mean	AUC sd	min fold	folds > 0.5	log loss	rank IC
lgbm	A_price	0.5375	0.0081	0.5281	7	0.6912	0.0559
lgbm	B_price_volume	0.5378	0.0077	0.5291	7	0.6912	0.0566
lgbm	C_plus_flow	0.5380	0.0071	0.5292	7	0.6911	0.0564
lgbm	D_plus_book	0.5391	0.0082	0.5296	7	0.6911	0.0592
lgbm	E_all	0.5388	0.0084	0.5284	7	0.6911	0.0586
logit	A_price	0.5344	0.0076	0.5227	7	0.6919	0.0514
logit	B_price_volume	0.5346	0.0081	0.5220	7	0.6919	0.0519
logit	C_plus_flow	0.5349	0.0069	0.5231	7	0.6918	0.0521
logit	D_plus_book	0.5368	0.0083	0.5246	7	0.6916	0.0557
logit	E_all	0.5367	0.0081	0.5244	7	0.6917	0.0554
rf	A_price	0.5369	0.0079	0.5262	7	0.6912	0.0549
rf	B_price_volume	0.5371	0.0079	0.5269	7	0.6912	0.0554
rf	C_plus_flow	0.5368	0.0079	0.5264	7	0.6912	0.0552
rf	D_plus_book	0.5381	0.0083	0.5268	7	0.6911	0.0573
rf	E_all	0.5381	0.0087	0.5265	7	0.6911	0.0572

Rule baselines scored on the same folds (15-minute horizon)

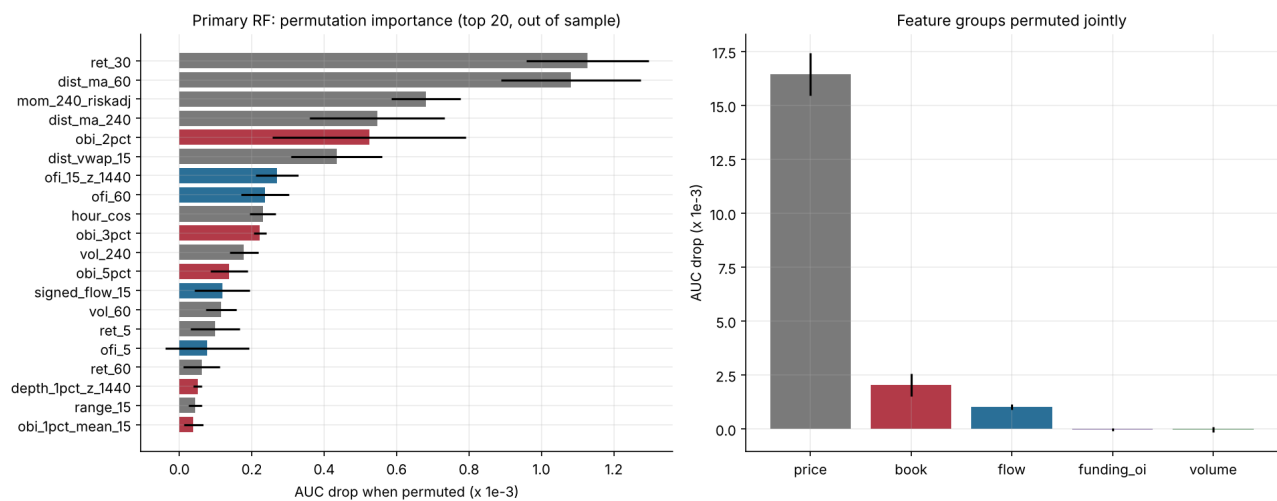
Rule	mean AUC	min	max
flow_ofi1	0.4925	0.4869	0.4988
flow_rule_ofi15z	0.4812	0.4700	0.4911
momentum_ret15	0.4745	0.4625	0.4857
prior_only	0.5000	0.5000	0.5000

Simple continuation rules are **anti-predictive** (AUC < 0.5): the market mean-reverts. The pre-declared rule (lowest validation log loss on set E) selected **rf_E_all_h15**. LightGBM was marginally better on AUC and marginally worse on log loss, a tie in practice.



Development out-of-fold: monthly AUC, hit rate, and the ML strategy's gross vs net return.

Interpretability (out of sample, fold F6)



Permutation importance of the primary random forest. Price features (distance from moving averages, recent returns) dominate. Book and flow add little; volume and funding/OI add essentially nothing. This is association, not causation.

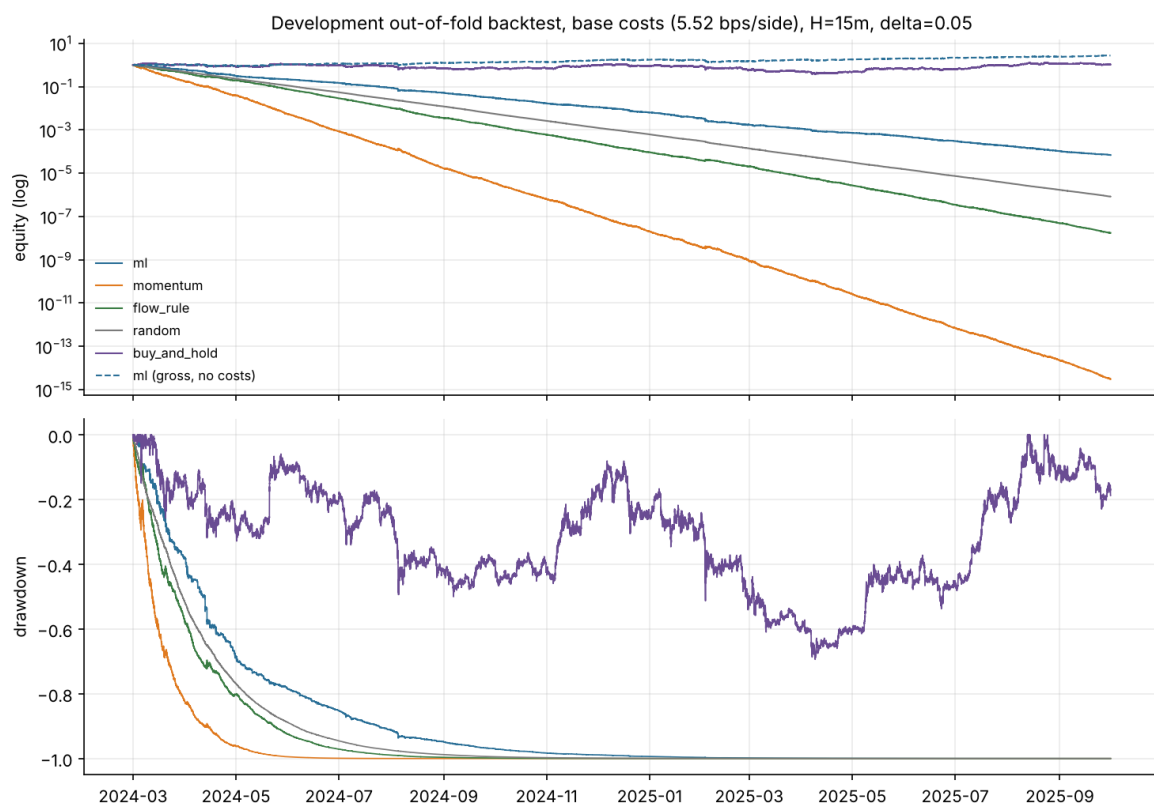
7. Signal threshold, development backtest and the freeze

Rule: LONG if $P(\text{up}) > 0.5 + \delta$, SHORT if $P(\text{up}) < 0.5 - \delta$, otherwise FLAT. δ is chosen on development out-of-fold predictions by the highest net Sharpe at base cost:

δ	share active	gross Sharpe	net Sharpe (base)	break-even bps/side	trades
0.005	0.910	1.274	-24.515	0.296	758,470
0.010	0.821	1.289	-24.270	0.301	684,381
0.020	0.650	1.295	-23.588	0.306	542,051
0.030	0.497	1.528	-22.118	0.370	414,528
0.050	0.261	2.092	-17.819	0.573	217,566

Exploratory (non-binding) check: even the most confident predictions never earn a round trip.

min $ P-0.5 $	share of minutes	mean gross trade (bps)	hit rate	round-trip cost (bps)
0.000	1.000	0.192	0.528	11.046
0.020	0.650	0.287	0.540	11.046
0.050	0.261	0.765	0.560	11.046
0.075	0.083	1.850	0.579	11.046
0.100	0.011	2.965	0.590	11.046
0.125	5.5e-05	-38.611	0.565	11.046



Development out-of-fold equity at base cost (log scale); dashed = ML before costs.

Freeze. docs/final_methodology.md (model rf_E_all_h15, $\delta = 0.05$, costs, execution, all features) was committed in 71ca5ad **before** the holdout was opened. Every holdout access is logged with the file's SHA-256: 2026-10-07T14:21:37+00:00 – final evaluation ETHUSD – f890b16b518e...; 2026-10-07T14:41:34+00:00 – replication evaluation SOLUSD – f890b16b518e.... The hashes match, so the methodology was not edited after the holdout was opened.

8. Final untouched holdout (2025-10-01 → 2026-09-30)

Prediction

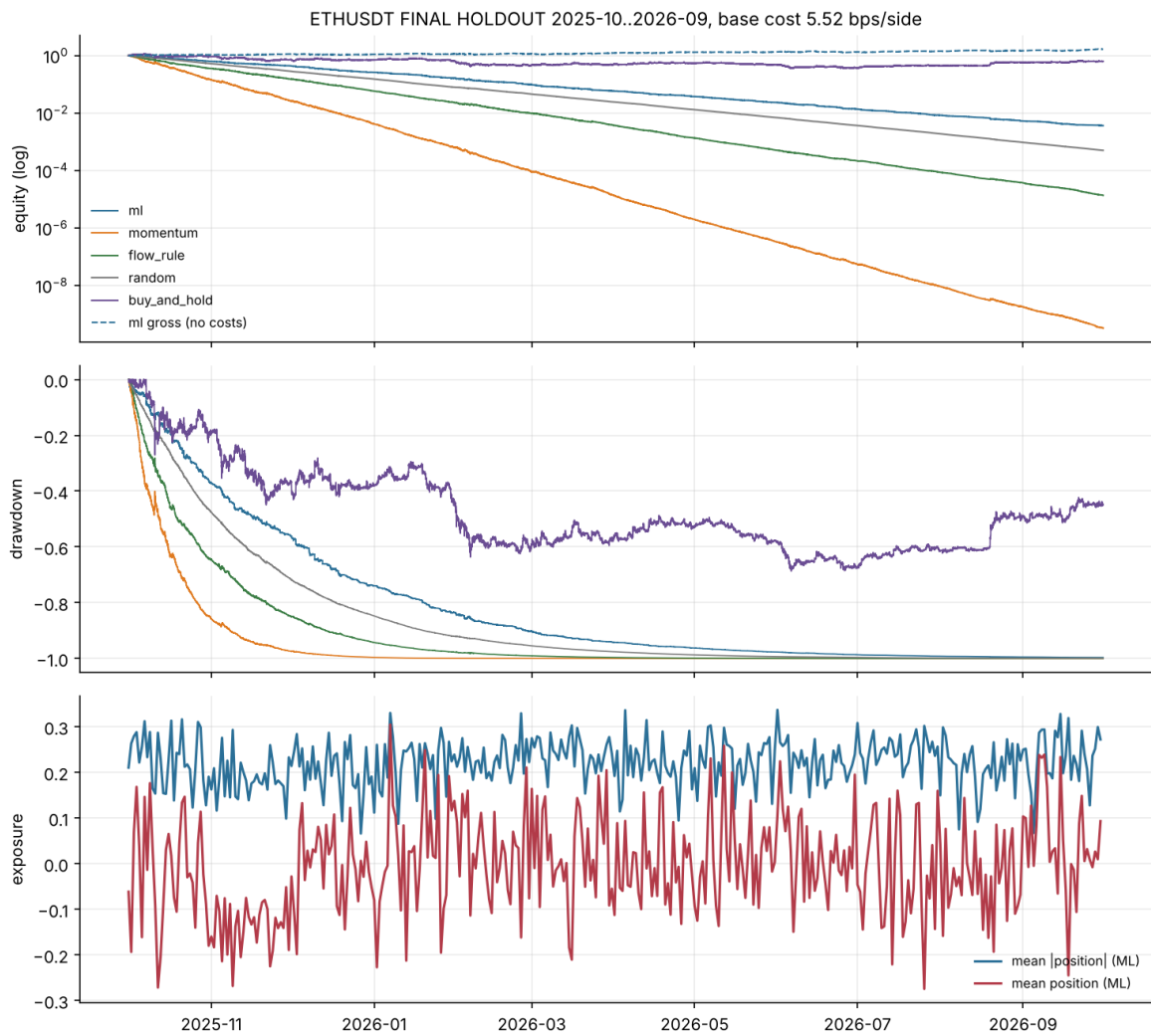
AUC	95% block-bootstrap CI	log loss / prior	ECE	rank IC	tradable AUC	acc / prec / rec / F1
0.5436	[0.5394, 0.5480]	0.6902 / 0.6931	0.0028	0.062	0.548	0.531 / 0.529 / 0.538 / 0.533

block	start	train rows	test rows	AUC	log loss	prior	IC
F1	2025-10-01	90,623	132,465	0.5363	0.6912	0.6931	0.0473
F2	2026-01-01	99,455	129,585	0.5488	0.6896	0.6931	0.0708
F3	2026-04-01	108,095	131,025	0.5491	0.6896	0.6930	0.0726
F4	2026-07-01	116,831	132,465	0.5412	0.6904	0.6931	0.0607

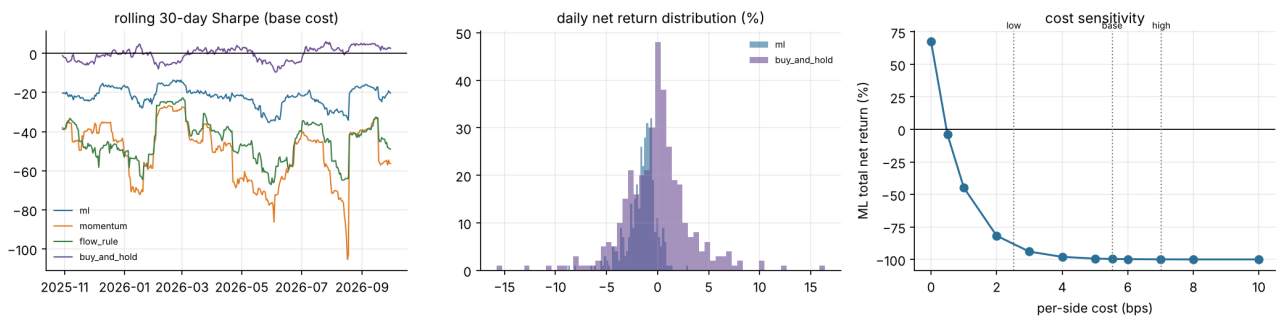
AUC is above 0.5 in every holdout block and month. 'Tradable AUC' scores the probability against the return that is actually capturable (VWAP of t+1 to VWAP of t+16), and it stays high. With 43 configurations tried, Bonferroni needs $z > 3.05$; the holdout AUC has $z = 19.6$.

Trading (frozen strategy and benchmarks)

strategy	cost	total return	Sharpe	Sortino	max DD	trades	win rate	avg trade (bps)	break-even bps/side
buy_and_hold	gross	-36.9%	-0.48	-0.67	-68.7%	1	-	-	-
flow_rule	gross	-51.9%	-2.81	-3.80	-53.2%	175,617	46.3%	-0.60	-0.37
ml	gross	67.5%	2.03	2.73	-12.8%	117,461	57.1%	0.70	0.49
momentum	gross	-78.1%	-3.18	-4.29	-81.4%	525,022	47.2%	-0.41	-0.39
random	gross	7.2%	0.91	1.43	-6.4%	117,194	49.9%	0.09	0.05
buy_and_hold	base	-36.9%	-0.48	-0.67	-68.7%	1	-	-	-
flow_rule	base	-100.0%	-41.43	-17.43	-100.0%	175,617	25.0%	-11.65	-0.37
ml	base	-99.6%	-20.37	-14.03	-99.6%	117,461	36.0%	-10.35	0.49
momentum	base	-100.0%	-44.11	-17.59	-100.0%	525,022	26.1%	-11.46	-0.39
random	base	-99.9%	-94.16	-18.72	-99.9%	117,194	28.1%	-10.95	0.05



Holdout equity (log scale) and drawdown at base cost for all strategies; dashed = ML before costs; bottom = ML exposure.



Rolling 30-day Sharpe, daily return distribution, and ML total return as a function of per-side cost.

9. Robustness

Feature ablation on the holdout (frozen procedure, 15 minutes)

model	set	AUC	tradable AUC	log loss	Sharpe gross	Sharpe net	net return	BE bps
lgbm	A_price	0.541	0.545	0.691	1.861	-15.279	-98.2%	0.591
lgbm	B_price_volume	0.541	0.545	0.691	1.578	-15.708	-98.3%	0.499
lgbm	C_plus_flow	0.541	0.545	0.691	2.108	-16.562	-98.5%	0.614
lgbm	D_plus_book	0.541	0.546	0.691	1.978	-18.184	-99.0%	0.530
lgbm	E_all	0.542	0.546	0.690	2.103	-18.790	-99.0%	0.548
logit	A_price	0.535	0.538	0.692	0.248	-7.061	-86.5%	0.180
logit	B_price_volume	0.534	0.537	0.692	0.278	-7.125	-86.7%	0.199
logit	C_plus_flow	0.536	0.541	0.691	0.893	-11.714	-96.2%	0.378
logit	D_plus_book	0.534	0.537	0.694	2.171	-8.941	-96.2%	1.047
logit	E_all	0.534	0.537	0.694	2.200	-8.868	-96.1%	1.066
rf	A_price	0.543	0.547	0.690	1.405	-19.770	-99.7%	0.367
rf	B_price_volume	0.542	0.547	0.690	1.386	-20.576	-99.6%	0.348
rf	C_plus_flow	0.543	0.547	0.690	1.408	-19.945	-99.6%	0.359
rf	D_plus_book	0.543	0.548	0.690	1.810	-20.690	-99.7%	0.440
rf	E_all	0.544	0.548	0.690	2.028	-20.372	-99.6%	0.493

Order flow and book bands do **not** add robust information beyond price on the holdout: a price-only RF reaches 0.543 against 0.544 with everything. For logistic regression the book sets lower AUC and push log loss above the prior (0.6931).

Horizons (holdout)

h	model	set	AUC	log loss	prior	IC
5	lgbm	A_price	0.5372	0.6910	0.6931	0.0569
5	lgbm	C_plus_flow	0.5367	0.6911	0.6931	0.0564
5	lgbm	E_all	0.5371	0.6910	0.6931	0.0574
5	logit	A_price	0.5337	0.6918	0.6931	0.0532
5	logit	C_plus_flow	0.5344	0.6914	0.6931	0.0527
5	logit	E_all	0.5306	0.6926	0.6931	0.0485
30	lgbm	A_price	0.5399	0.6908	0.6931	0.0566
30	lgbm	C_plus_flow	0.5388	0.6910	0.6931	0.0549
30	lgbm	E_all	0.5400	0.6908	0.6931	0.0554
30	logit	A_price	0.5316	0.6920	0.6931	0.0459
30	logit	C_plus_flow	0.5328	0.6917	0.6931	0.0452
30	logit	E_all	0.5303	0.6948	0.6931	0.0434
60	lgbm	A_price	0.5372	0.6914	0.6931	0.0530
60	lgbm	C_plus_flow	0.5357	0.6916	0.6931	0.0493
60	lgbm	E_all	0.5370	0.6913	0.6931	0.0539
60	logit	A_price	0.5294	0.6923	0.6931	0.0412
60	logit	C_plus_flow	0.5304	0.6921	0.6931	0.0397
60	logit	E_all	0.5255	0.6954	0.6931	0.0351

Volatility regimes (holdout; cut-offs fixed on 2023-03 → 2024-02)

regime	cost	minutes	AUC	sum of bar returns	turnover
1_low	base	76,923	0.550	-0.922	1,458
2_mid	base	126,901	0.547	-1.356	2,759
3_high	base	321,776	0.541	-3.304	6,881
1_low	gross	76,923	0.550	-0.117	1,458
2_mid	gross	126,901	0.547	0.168	2,759
3_high	gross	321,776	0.541	0.496	6,881

Confirmatory test of the main development finding on the holdout

signal	h	n	β (bps / sd)	HAC t	p	Holm p
ofi_1	5	525,595	-0.0690	-2.7360	0.0062	0.0373
ofi_1	15	525,585	-0.0551	-1.1936	0.2326	0.2326
obi_1pct	5	521,401	0.1348	2.0274	0.0426	0.1705
obi_1pct	15	521,391	0.3116	1.5826	0.1135	0.2270
ofi_15	5	525,595	-0.1005	-2.1306	0.0331	0.1656
ofi_15	15	525,585	-0.2289	-1.8841	0.0596	0.1787

Replication on SOLUSDT (frozen settings, nothing re-tuned)

Holdout AUC 0.5343, tradable AUC 0.5373. ML strategy:

cost	total return	Sharpe	max DD	break-even bps/side
gross	43.6%	1.74	-25.6%	0.89
low	-53.3%	-3.18	-55.7%	0.89
base	-87.7%	-8.78	-87.8%	0.89
high	-93.7%	-11.41	-93.7%	0.89

Multiple testing and data snooping

- Tried in development: 43 model configurations, 5 thresholds, 36 univariate tests (Holm/BH adjusted), controls, deciles and events.
- Holdout AUC passes Bonferroni over 43 configurations ($z = 19.6$ vs 3.05 needed): the predictability is not a selection artefact.
- Gross Sharpe 2.03 (CI 0.32–3.75) is below the ≈ 3.3 expected maximum of 215 zero-skill variants, so even the gross profit claim is weak.
- Net Sharpe -20.4 (CI -23.0 to -18.3) is unambiguously negative.

10. Conclusions

Question	Answer (evidence)
Is there order-flow predictability?	Yes statistically: 1-min OFI predicts reversal (Holm p down to 7e-21 in development; p = 0.037 at 5 min on the holdout). No economically: ≈ 0.07 – 0.24 bps per sd.
Does order flow add value beyond price/volume?	Marginally in development ($\leq +0.002$ AUC), not robustly on the holdout.
Does ML improve the signal?	Yes as a forecaster (AUC ≈ 0.54 vs < 0.5 for rules); trees beat logit. No economic viability.
Does it survive out of sample?	Yes: AUC 0.5436 [0.5394, 0.5480], 12/12 months > 0.5 , calibrated.
Does it survive realistic costs?	No: break-even 0.49 bps per side vs 5.52 base (2.52 even at low cost); net -99.6%.
Is it stable over time and regimes?	Prediction yes: no decay over 31 out-of-sample months; AUC 0.541–0.550 across volatility terciles. Net P&L; negative everywhere.
Does it replicate on SOL?	Yes qualitatively: holdout AUC 0.534; also untradable (break-even 0.89 bps per side).
Strongest limitations	Taker-only cost model; percentage-band book (no touch, no queue); single venue; 1-minute resolution; slippage assumed.

Bottom line. One-minute ETH perpetual price paths mean-revert in a small, stable and statistically robust way, and taker order flow is mildly informative about that reversal. The predictable component is worth about half a basis point per side, roughly one tenth of the cost of taking liquidity. **Predictive accuracy $\neq$ trading profitability.** A viable version would need maker execution and queue-level data, which this public dataset cannot support.

Reproduction

pip install -r requirements.txt; pytest -q (30 tests); then run scripts/run_stage2.py → measure_spread.py → build_dataset.py → leakage_audit.py → run_eda_stats.py → run_models.py → run_dev_backtest.py → run_interpret.py → freeze_methodology.py → run_holdout.py → run_replication.py → run_significance.py → make_results_table.py → make_notebooks.py → make_pdf_report.py. Raw data are rebuilt from the public archive; the manifest records each file's URL and SHA-256.